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Milk fat as a structuring agent of plastic lipid bases

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ABSTRACT

The global legislation regarding the elimination of trans fat in foods has led to the need for technological solutions to produce plastic fats. Currently, the industrial method used to obtain lipid bases with different physical properties is the chemical interesterification of blends from hardfats and vegetable oils. Milk fat can be an alternative to this process, which is natural plastic fat, as a structurant to obtain plastic lipid bases containing vegetable oils. This study aimed to evaluate the ability of the anhydrous milk fat (AMF) to structure lipid bases made from AMF and high oleic sunflower oil (HOSO) (~80% oleic acid) blends. The blends were prepared in the following AMF:HOSO proportions (% w/w): 100:00 (control); 90:10; 80:20; 70:30; 60:40; and 50:50, and characterized for fatty acid and triacylglycerol composition, solid fat content, compatibility, melting point, thermal behavior of crystallization and melting, crystallization kinetics, microstructure, polymorphism and hardness. All blends showed compatibility between the constituents, which is fundamental for the stability of plastic fats. The anhydrous milk fat governed the crystallization of these lipid bases, presenting crystals of < 30 μm of diameter, crystallized in polymorphic form β′. The blends 70:30, 60:40 and 50:50 AMF:HOSO exhibited suitable profile for technological applications in the food industry, characterized by a solid fat content lower than 32% at 10 °C, and higher than 10% at 21 °C, and melting point near the body temperature.

1. Introduction

The global demand for fat for food uses increased from 52 to 104 million tons between 1980 and 2000 and is expected to reach 184 million tons until 2020 (Lands, 2005). An adequate plasticity is desired for a technical fat or tailor-made fat to be used in the food industry (Martin, Carapelli, Visantainer, Matsushita, & Souza, 2005). The term plastic fat is used to define fats that have the ability to trap the fat liquid phase in the crystal lattice. This ability is governed by the solid-to-liquid fat ratio in the system at a given temperature (Karabulut, Turan, & Ergin, 2004). The fat crystallization and the consequent entrapment of liquid fat in the crystal network can be affected by several factors, including chemical composition of the lipid matrix, crystallization kinetics, crystal structure, stability, and processing conditions (Truong, Morgan, Bansal, Palmer, & Bhandari, 2015).

In the old days, plastic fats for industrial applications were obtained through the partial hydrogenation of vegetable oils (O'Brien, 2009). This process, however, leads to the formation of *trans* fatty acids, such as the elaidic acid (C18:1 *trans*-9), which is associated with risk factors for coronary disease and different metabolic and functional disorders (Mozaffarian, Katan, Ascherio, Stampfer, & Willett, 2006). Hardfats, fat materials obtained from the total hydrogenation of different vegetable oils, have emerged as a technological solution for the elimination of *trans* fat in processed foods. Alone, hardfats have limited technological applications due to their homogeneous chemical composition and high melting point (Ribeiro, Basso, & Kieckbusch, 2013). When mixed with vegetable oils, they result in plastic lipid bases due to their modulating effect on crystallization and the recognized ability to entrap the oil in the crystal network. However, several authors (Grimaldi, Gonçalves, Gioielli, & Simões, 2001; Ribeiro, Grimaldi, Gioielli, & Gonçalves, 2009; Guedes et al., 2014) have reported that, in general, the simple mixing of hardfats with different vegetable oils can lead to an incompatibility due to the drastic difference in the molecular size and polymorphic habit of triacylglycerols. Thus, the method currently used by the industry to obtain compatible lipid bases is chemical interesterification of blends hardfats and vegetable oils, which promotes a rearrangement of triacylglycerols, making it possible to obtain plastic lipid bases with less content or free of *trans* isomers (Ribeiro, Basso, Grimaldi, Gioielli, & Gonçalves, 2009). However, interesterification is a costly process when compared to partial hydrogenation (Mba, Dumont, & Ngadi, 2015),

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besides involving high consumption of water and energy (Patterson, 2010), and the need for further an additional stage of clarification and deodorization of the interesterified material. In addition, the interesterified lipid base can have lower oxidative stability and presence of monoacylglycerols and diacylglycerols, which delay the fat crystallization (Smith, Bhaggan, Talbot, & van Malssen, 2011).

In this context, milk fat, wich is a natural plastic fat, may be a promising technological alternative for the production of new lipid bases. This natural plasticity is due to the milk fat composition, which contains > 400 fatty acids participating in the structure of triacylglycerols with different molecular masses and a wide range of crystallization and melting temperatures. Milk fat is completely liquid above 40 °C and completely solid below −40 °C. Between these extremes of temperature it exists as a mixture of liquid and crystallized fat, being capable of governed the crystallization of lipid bases, giving them structure and softness as a function of the β′ polymorphic form (MacGibbon & Taylor, 2006; Ramel, Peyronel, & Marangoni, 2016). Its structuring capacity was demonstrated in plastic lipid bases, interesterified or not, obtained from the mixture of milk fat and canola oil (Rousseau, Forestiere, Hill, & Marangoni, 1996), maize oil (Rodrigues, Anton, & Gioielli, 2003), sunflower oil (Rodrigues-Ract, Otting, Poltronieri, Silva, & Gioielli, 2010), and palm oil (Danthine, 2012).

In this context, the objective of this study was evaluate the ability of anhydrous milk fat (AMF) structure plastic lipid bases with a reduced content of saturated fatty acids, without the use of chemical modification. Our hypothesis is that blends from milk fat and high oleic sunflower oil (~80% C18:1 *cis*-9), with higher oxidative stability can result in specific plastic lipid systems, stabilized by crystallized milk fat containing HOSO trapped in the crystal lattice for different technological applications.

2. Material and methods

2.1. Raw materials

Anhydrous milk fat (Fonterra Ltda., Brazil) was in accordance with the parameters established by the Brazilian legislation (Brasil, 1996), with a fat level of 99.8% and moisture and acidity (% oleic acid) levels of 0.2% and 0.38%, respectively. The HOSO (Cargill Foods, Brazil) was in compliance with the Brazilian legislation, which acidity (% oleic acid) content of 0.08%, iodine value 87.45 g I_2/100 g, and saponification index 190.8 mg KOH/g (Brasil, 2005; Brasil, 2006).

2.2. Preparation and characterization of blends

Lipid bases (250 g) were made from blends of anhydrous milk fat and high oleic sunflower oil (AMF:HOSO % w/w) in the following proportions 100:00 (control); 90:10; 80:20; 70:30; 60:40; and 50:50, in triplicate, after complete melting of the raw materials at 45 °C for 60 min. The blends were characterized for the fatty acid and triacylglycerol composition, solid fat content, compatibility, melting point, thermal behavior, crystallization kinetics microstructure, polymorphism, and hardness. The blends were stored at 3 °C in an incubator chamber (Marconi, MA415/S) until the time of analysis. Prior to each analytical determination, the lipid bases were melted at 45 °C for 60 min to erase crystal memory. They were then stabilized according to the protocols required by each determination.

2.2.1. Fatty acid composition

The AMF and the AMF:HOSO blends were dissolved in hexane and methyl acetate and converted to fatty acid methyl esters (FAME) using freshly prepared sodium methoxide in methanol according to Christie (1982). The mixture was neutralized with oxalic acid, and calcium chloride was added to remove methanol residues. FAME were separated and quantified in 1.0 μL sample at a split ratio of 1:50 using a gas chromatograph (GC) (model 7820A, Agilent Technologies) fitted with a flame-ionization detector and equipped with a CP-Sil 88 fused-silica capillary column (100 m × 0.25 mm × 0.2 μm film thickness; Varian Inc.). The operating conditions were performed as described by Cruz-Hernandez et al. (2007), including injector and detector temperatures of 250 °C, hydrogen as carrier gas (1 mL/min), and nitrogen as makeup gas (30 mL/min) and purified air (286 mL/min) for the flame-ionization detector (35 mL/min). The initial temperature was 45 °C for 4 min, then increased to 175 °C at a ratio of 13 °C/min, and maintained for 27 min, and increased to 215 °C at 4 °C/min and maintained for 35 min. The FAME were identified by comparison of retention times with reference FAME standards (Sigma-Aldrich®, Nu-Chek Prep, Inc., and Larodan Fine Chemicals), and according to their order of elution under similar GC operation conditions (Cruz-Hernandez et al., 2007; Kramer, Hernandez, Cruz-Hernandez, Kraft, & Dugan, 2008). The fatty acid composition was expressed as the percentage of total fatty acids using theoretical response factors (Wolff, Bayard, & Fabein, 1995).

For the HOSO, the fatty acids were subjected to esterification according to the method of Hartman and Lago (1973). The FAME (1.0 μL sample) at a split ratio of 1:50 were separated (AOCS, 2009; method Ce 1f-96) and quantified using a gas chromatograph (model 6850, Agilent Technologies) fitted with a flame-ionization detector and equipped with a DB-23 capillary column (50% Cyanopropyl-methylpolysiloxane; 60 m × 0.25 mm × 0.25 μm film thickness; Agilent Technologies). The operating conditions were injector temperature 250 °C, detector temperature 280 °C, helium as a carrier gas, column flow of 1.0 mL/min, linear velocity 24 cm/s. The initial oven temperature was 110 °C, which was maintained for 5 min, followed by a heating ramp of 5 °C/min to 215 °C, which was maintained for 24 min. FAME qualitative composition was determined by comparing the peak retention times with C4-C24 FAMEs reference standard (Sigma-Aldrich®, Brazil), while the quantitative composition was determined by area normalization, expressed as the percentage of mass (g/100 g of total fatty acids).

2.2.2. Triacylglycerol composition

The triacylglycerol composition of both the AMF and HOSO was analyzed in a gas chromatograph (model 6850, Agilent Technologies) fitted with a flame-ionization detector and equipped with a DB-17 capillary column (50% Phenyl-methylpolysiloxane; 30 m × 0.25 mm × 0.15 μm film thickness; Agilent Technologies). The operation conditions were: split injection, ratio 1:100; column temperature: 250 °C, programmed up to 350 °C at 5 °C/min; carrier gas: helium, at 1.0 mL/min flow rate; injector temperature: 360 °C; detector temperature: 375 °C; injection volume: 1.0 μL; sample concentration: 5 mg/5 mL of tetrahydrofuran. The composition of triacylglycerolspecies in HOSO was determined from the fatty acid composition obtained experimentally. The calculations were performed by software PR Oils based on the regiospecific distribution of fatty acids at the *sn*-1,3 and *sn*-2 positions (Antoniosi Filho, Mendes, & Lanças, 1995). Triacylglycerol groups from the AMF and HOSO were identified after integration of the chromatogram peaks and expressed as the percentage by mass (g/100 g of total triacylglycerols). The triacylglycerol composition of the AMF:HOSO blends was calculated from the chemical composition of the raw materials, considering that the triacylglycerol composition of the blends represents a linear combination of AMF and HOSO in the blends.

2.2.3. Solid fat content, compatibility and melting point of blends

The solid fat content (SFC) was determined according to the AOCS method Cd 16b-93 (2009) by conditioning the samples at reading temperatures, using nuclear magnetic resonance spectrometer (NMR) (Bruker pc120 Minispec, Germany) and high-accuracy dry bath (0–70 °C ± 0.1 °C) (TCON 2000 - Duratech, USA). After melting (45 °C/60 min.), the samples were subjected to tempering procedures (60 °C/5 min), followed by stabilization for 1 h at 0 °C. The determinations were carried out in series at temperatures of 10, 15, 20, 25, 30, 35, and 40 °C for the unstabilized fats.

From those data, the compatibility of the blends was evaluated through the compatibility diagram, by correlating the HOSO ratio and

the solid fat content at 10, 15, 20, 25, and 30 °C (Quast, Luccas, Ribeiro, Cardoso, & Kieckbusch, 2013).

The melting point was calculated for the temperature corresponding to 4% solids content, obtained from the SFC curve given by NMR (Karabulut et al., 2004).

2.2.4. Thermal behavior

The thermal analyses of the AMF and AMF:HOSO blends were performed by differential scanning calorimetry (DSC) (DSC Q2000 - TA Instruments, USA) using a calibration factor determined with indium, according to the methodology of AOCS Cj 1-94 (2009). After melting (45 °C/60 min), 10 mg sample was weighed on aluminum pans, using an empty, hermetically sealed aluminum pan as a baseline reference. For the crystallization behavior, the analysis conditions were: isotherm at 80 °C for 10 min followed by cooling at a rate of 2 °C/min to −80 °C; for the melting behavior, the sample was maintained at −80 °C for 30 min, followed by a heating rate of 5 °C/min to 80 °C. Then, the parameters onset temperature (T_o), final temperature (T_{end}), peak temperature (T_p) and enthalpy (ΔH) of crystallization and melting were determined (Biliaderis, 1983).

2.2.5. Crystallization kinetics

The blends were subjected to isothermal crystallization at 15 °C in a nuclear magnetic resonance spectrometer (NMR) equipment (Bruker pc120 Minispec, Germany), using high precision dry baths (0–70 °C ± 0.1 °C) (TCON 2000 - Duratech, EUA) and Lauda circulator heater (E200 Ecoline-star edition). The AMF and AMF:HOSO blends were melted (45 °C/60 min) and maintained at 70 °C for 60 min for the erase crystal memory (Ribeiro et al., 2009). The crystallization kinetics was characterized according to the non-linearized Avrami equation, based on the induction period, maximum solids content, and the crystallization stabilization time (Campos, 2005).

$$\frac{SFC(t)}{SFC_{máx}} = 1 - e^{-kt^n} \quad (1)$$

where SFC(t) is the solid fat content (%) as a function of time (t), SFC_{max} is the limit solid fat content, k is the Avrami constant (min^{-1}), which considers both nucleation and growth rate, and n is the Avrami exponent, which indicates the crystal growth mechanism (Wright, Hartel, Narine, & Marangoni, 2000).

2.2.6. Microstructure

The crystal morphology was determined using a polarized light microscopy (Olympus BX51, USA), coupled to a digital camera (Media Cybernetic, USA). The AMF and AMF:HOSO blends were melted (45 °C/60 min), and 10 uL was placed on a preheated glass slide (80 °C/15 min) with the aid of a capillary tube, which was covered with a cover slip. The slides were maintained in an incubator (Marconi, MA415/S) at 15 °C and 30 °C for 24 h. Three images were captured using the Image-Pro Plus software version 7.0 (Media Cybernetic, USA), with polarized light and 20× magnification.

The quantitative analysis was performed at 15 °C, using the following parameters: total number of crystals found in the image, crystal density, crystal diameter, percentage of agglomerated crystals, and diameter of the non-agglomerated crystals (Gamboa & Gioielli, 2006).

2.2.7. Polymorphism

The AMF and AMF:HOSO blends were melted (45 °C/60 min) and the polymorphic form of the fat crystals was determined after 7 and 50 days of stabilization at 15 °C by X-ray diffraction (XRD), according to the methodology AOCS Cj 2-95 (AOCS, 2009). The analyses were carried out in a Philips PW 1710 diffractometer (PANalytical, The Netherlands), using Bragg-Brentano geometry (θ:2θ) with Cu kα radiation (λ = 1.54056 Å, 40 KV voltage and 30 mA current). Measurements were obtained with step of 0.02° in 2θ and an acquisition time of 2 s, with scans of 15 to 30° (2θ scale). The identification of the polymorphic shape was determined from the characteristic interplanar distances or short spacings of each type of crystal (Schenk & Peschar, 2004).

2.2.8. Hardness

Hardness was determined using a Texture Analyzer (TA.XT Plus-Stable Micro Systems, England) using an acrylic cone with an angle of 45° and no truncation. The AMF and AMF:HOSO blends were melted (45 °C/60 min) in 50 mL beaker and incubated at 5 °C for 24 h in an incubator chamber (Marconi, MA415/S) for fat crystallization, and then for 24 h at 15 °C (reading temperature). The tests were performed in quadruplicate using the following parameters: distance = 10 mm; velocity = 2 mm/s; time = 5 s (Queirós, Grimaldi, & Gigante, 2016).

2.3. Statistical analysis

The effect of the addition of high oleic sunflower oil (HOSO) to anhydrous milk fat (AMF) on the solid fat content, melting point, crystallization kinetics and hardness of the lipid bases was evaluated by Analysis of Variance (ANOVA). In case of difference, the averages were compared by the Tukey's test at a level of significance of 5%. The analyses were performed using STATISTICA 7.0 software (StatSoft Inc., USA). The effect of the solid fat content on the hardness of the lipid bases was assessed by correlation analysis at 5% level of significance, using EXCEL software 16.0 (2016).

3. Results and discussion

3.1. Fatty acid and triacylglycerol composition

The milk fat presented typical composition (Table 1), with predominance of myristic (C14:0), palmitic (C16:0) and stearic acid (C18:0) as a saturated fatty acids, and oleic acid (C18:1 *cis*-9) among the unsaturated fatty acids. High oleic sunflower oil contained 75.73 g/100 g oleic acid (C18:1 *cis*-9) and 12.06 g/100 g linoleic acid (C18:2 n-6), as recommended for a sunflower oil to be classified as high oleic acid oil in Brazil (Brasil, 2006). The addition of 50% HOSO to AMF resulted in a lipid base with lower contents of saturated fatty acids, which are related to the increased plasma total cholesterol and low density lipoproteins (LDL), and higher contents of unsaturated fatty acids related to the reduction of plasma LDL (Allman-Farinelli, Gomes, Favaloro, & Petocz, 2005; Parodi, 2009). A reduction of 42.23% and 45.75% of myristic (C14:0) and palmitic acid (C16:0) and increase of 145.26% in oleic acid was observed in the blend containing 50% HOSO.

Among the low molecular mass (C26-C42) triacylglycerols (TAGs) (Table 2), the AMF showed higher concentrations of composite species with 36, 38, 40, and 42 carbon atoms. With respect to the high molecular mass TAGs (C44-C60), higher concentrations of species formed by 44 to 52 carbon atoms were observed when compared to TAGs containing 54 carbon atoms. The HOSO, presented 14 TAGs species containing 50 to 60 carbon atoms, the majority containing 54 carbon atoms (83.3 g/100 g of total TAGs), especially of the OOO and OOL (O = oleic acid, L = linoleic acid). The addition of HOSO to the AMF led to a reduction in the low molecular weight TAGs and an increase in high molecular mass TAGs, especially those containing 54 carbon atoms, which increased from 1.80 g/100 g to 42.37 g/100 g in the blend with 50% HOSO. A reduction of 41.32% in low molecular mass TAGs and 71.78% increase in high molecular mass TAGs was observed in the 50:50 blend.

3.2. Solid fat content, compatibility, and melting point of the blends

A significant reduction of the solid fat content was observed with the increase in temperature (Fig. 1a), with a greater reduction of 15–20 °C for all blends, since it is the melting temperature of the major milk fat TAGs (MacGibbon & Taylor, 2006). The reduction of the solid fat content of the blends was less marked with the higher HOSO

Table 1
Fatty acid composition (g/100 g of total fatty acids) of anhydrous milk fat (100:00), high oleic sunflower oil (HOSO) and their blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w) (n = 1).

Fatty acid	AMF:HOSO (% w/w)						
	100:00	90:10	80:20	70:30	60:40	50:50	HOSO
C4:0	3.59	3.17	2.88	2.53	2.30	1.78	–
C5:0	0.03	0.02	0.02	0.02	0.02	0.02	–
C6:0	2.25	2.13	1.98	1.70	1.54	1.17	–
C7:0	0.02	0.02	0.02	0.02	0.01	0.01	–
C8:0	1.37	0.05	0.05	0.05	0.05	0.77	–
C9:0	0.03	0.03	0.03	0.02	0.02	0.02	–
C10:0	2.93	3.17	2.97	2.51	2.26	1.73	–
C10:1*cis*-9	0.27	0.26	0.25	0.20	0.18	0.13	–
C11:0	0.08	0.08	0.07	0.06	0.05	0.04	–
C12:0	3.46	4.37	4.05	3.43	3.05	2.38	–
C12:1 *cis*-9 + C13:0	0.20	0.21	0.19	0.16	0.14	0.11	–
C14:0 iso	0.11	0.10	0.09	0.07	0.06	0.06	–
C14:0	11.08	11.83	10.75	9.20	8.02	6.40	0.11
C15:0 *iso*	0.28	0.30	0.27	0.23	0.20	0.16	–
C15:0 *anteiso*	0.53	0.64	0.57	0.50	0.44	0.35	–
C14:1 *cis*-9	1.02	0.94	0.85	0.74	0.65	0.51	–
C15:0	1.14	1.28	1.15	0.99	0.87	0.70	–
C16:0 *iso*	0.28	0.25	0.23	0.19	0.16	0.14	–
C16:0	31.30	28.62	25.82	22.89	20.11	16.98	6.05
C16:1 *trans*-9 + C17:0 *iso*	0.40	0.49	0.43	0.38	0.32	0.26	–
C16:1 *trans*-12	0.18	0.18	0.17	0.15	0.14	0.10	–
C17:0 *anteiso*	0.42	0.40	0.37	0.33	0.30	0.22	–
C16:1 *cis*-9	1.44	1.22	1.07	0.95	0.80	0.70	0.16
C17:0	0.70	0.69	0.61	0.53	0.46	0.39	0.05
C18:0 *iso*	0.06	0.04	0.04	0.03	0.03	0.02	–
C17:1 *cis*-9	0.23	0.20	0.17	0.17	0.15	0.12	0.06
C18:0	10.01	8.54	7.75	7.15	6.42	5.89	3.67
C18:1 *trans*-4	0.03	0.03	0.02	0.03	0.02	0.02	–
C18:1 *trans*-5	0.01	0.02	0.01	0.02	0.01	0.01	–
C18:1 *trans*-6,-7,-8	0.23	0.13	0.12	0.10	0.09	0.08	–
C18:1 *trans*-9	0.22	0.14	0.12	0.11	0.09	0.08	–
C18:1 *trans*-10	0.33	0.18	0.15	0.13	0.12	0.13	–
C18: 1 *trans*-11	1.14	2.53	2.21	1.94	1.63	1.32	–
C18:1 *trans*-12	0.30	0.26	0.23	0.20	0.17	0.14	–
C18:1 *trans*-13,-14	0.48	0.54	0.29	0.25	0.21	0.17	–
C18: 1 *cis*-9	19.20	21.18	27.47	34.32	40.23	47.09	75.73
C18:1 *cis*-11	0.76	0.68	0.75	0.90	1.03	1.15	–
C18:1 *cis*-12	0.25	0.11	0.12	0.13	0.14	0.16	–
C18:1 *cis*-13	0.10	0.07	0.07	0.07	0.08	0.08	–
C18:1 *trans*-16	0.26	0.31	0.27	0.25	0.22	0.20	–
C19:0	0.12	0.15	0.13	0.12	0.14	0.13	–
C18:1 *cis*-15	0.06	0.11	0.09	0.08	0.07	0.07	–
C18:2 *trans*-9, 12	0.01	0.02	0.01	0.01	0.01	0.02	–
C18: 2 *cis*-9,*trans*-12	0.06	0.09	0.08	0.07	0.06	0.05	–
C18:2 *trans*-9,*cis*-12	0.06	0.05	0.04	0.04	0.03	0.03	–
C18:2 n-6	1.63	1.80	2.74	3.78	4.72	5.81	12.06
C20:0	0.15	0.16	0.16	0,17	0,18	0.20	0.32
C18:3 n-6	0.02	0.01	0.01	0.01	0.01	0.01	0.40
C18:3 n-3 + C20:1 *c*-11	0.23	0.60	0.57	0.56	0.53	0.50	–
CLA *cis*-9, *trans*-11	0.53	1.05	0.91	0.81	0.68	0.55	–
CLA *trans*-9, *cis*-11	0.02	0.04	0.02	0.03	0.03	0.02	–
CLA *trans*-10, *cis*-12	0.01	0.01	0.01	0.00	0.01	0.00	–
C21:0	0.04	0.04	0.04	0.03	0.03	0.03	0.28
C20:2 n-6	0.03	0.03	0.02	0.02	0.02	0.02	–
C22:0	0.05	0.16	0.23	0.31	0.37	0.48	0.84
C20:3 n-6	0.07	0.03	0.03	0.02	0.02	0.01	–
C20: 4 n-6	0.13	0.04	0.03	0.03	0.03	0.02	–
C23:0	0.01	0.04	0.04	0.04	0.03	0.03	–
C20:5 n-3	0.02	0.07	0.06	0.06	0.05	0.03	–
C24:0	0.04	0.07	0.09	0.12	0.14	0.17	0.28
C22:5 n-3	0.01	0.04	0.04	0.03	0.03	0.03	–
Σ Saturated	70.06	66.35	60.39	53.24	47.26	40.27	11.32
Σ Unsaturated	29.94	33.65	39.61	46.76	52.74	59.73	88.68

concentration, due to the increase in the concentration of low melting triacylglycerols. This difference was 17.22% for AMF and 6.82% for the blend 50:50 AMF:HOSO. At all temperatures, except 35 °C, the addition of HOSO to the AMF resulted in a significant reduction of the solid fat content. At 35 °C, all blends differed significantly from the AMF, with no significant differences among each other (shown in the Supplementary material, Table TS1).

Table 2
Triacylglycerol composition (g/100 g of total triacylglycerols) of anhydrous milk fat (100:00), high oleic sunflower oil (HOSO), and their blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w) (n = 1).

TAGs	AMF:HOSO (% w/w)						HOSO
	100:00	90:10	80:20	70:30	60:40	50:50	
C26	0.37	0.33	0.29	0.26	0.22	0.19	–
C28	1.96	1.76	1.57	1.37	1.18	0.98	–
C30	1.76	1.58	1.40	1.23	1.06	0.88	–
C32	3.92	3.52	3.13	2.74	2.35	1.96	–
C34	6.67	6.00	5.33	4.67	4.00	3.34	–
C36	12.00	10.81	9.61	8.40	7.20	5.99	–
C38	16.03	14.44	12.83	11.23	9.62	8.01	–
C40	12.16	10.95	9.73	8.51	7.29	6.07	–
C42	8.58	7.73	6.87	6.00	5.14	4.28	–
C44	7.43	6.68	5.94	5.20	4.45	3.71	–
C46	7.64	6.87	6.11	5.34	4.58	3.82	–
C48	7.32	6.58	5.85	5.12	4.39	3.66	–
C50	7.06	6.40	5.75	5.11	4.46	3.83	0.59
C52	5.30	6.12	6.95	7.76	8.58	9.50	13.52
C54	1.80	9.93	18.05	26.17	34.31	42.37	83.03
C56	–	0.12	0.25	0.36	0.48	0.60	1.21
C58	–	0.10	0.20	0.31	0.41	0.52	1.03
C60	–	0.06	0.12	0.19	0.25	0.31	0.62
Σ C26-C42[a]	63.45	65.03	58.58	52.05	43.77	37.23	0.00
Σ C44-C60[b]	36.54	34.97	41.42	47.95	56.23	62.77	100.00

[a] Σ C26-C42: low molecular mass.
[b] Σ C44-C60: high molecular mass.

The application potential of lipid bases depends on the solid fat content at different temperatures. The 60:40 and 50:50 AMF:HOSO blends that had solid fat content < 32% at 10 °C can be used in refrigerated products such as margarine and spreads for the ease of spreadability under refrigeration. All lipid bases, except the 50:50 AMF:HOSO blend, may be classified as fats resistant to oil exudation, because they contain > 10% solid fat at 20 °C (Lida & Ali, 1998).

The 80:20 and 70:30 AMF:HOSO blends with solid fat contents ranging from 15 to 20% at 20 °C presented a suitable solid profile and plasticity for use in baking and in technical fats for confectionery. According to Wassel & Young (2007), shortenings for these purposes should produce stable emulsions with good cream-forming properties. All blends of the present study are capable of conferring structure to lipid-based products with melting characteristics appropriate to body temperature, since they have crystallized fat above 25 °C, wich is correlated to the specific hardness of a lipid base (Karabulut et al., 2004).

The evaluation of compatibility of the blends (Fig. 1b), showed a linear evolution in all temperatures studied, which characterizes full compatibility between AMF and HOSO (Timms, 1984). In daily practice, this compatibility is due to the solubility in the solid state of the triacylglycerol molecules of the blends, suggesting a co-crystallization of lipid blends, which characterizes the stability of plastic fats (Braipson-Danthine & Deroanne, 2003). Incompatible lipid systems result in the formation of eutectic and monotectic solutions, not suitable for technological applications of plastic fats. When a complex fat such as milk fat is used as a structuring agent, compatible and incompatible systems can be formed. While blends made from milk fat and canola oil (Rousseau, Forestiere, et al., 1996) and milk fat and hydrogenated cotton oil (Shen, Birkett, Augustin, Dungey, & Versteeg, 2001) resulted in compatible systems, blends made from milk fat and corn oil (Rodrigues & Gioielli, 2003), milk fat and phytosterol esters (Rodrigues, Torres, Mancini-Filho, & Gioielli, 2007) and milk fat and palm oil (Danthine, 2012) presented different degrees of incompatibility, which hinders its technological application.

A significant reduction of melting point was observed in the fat

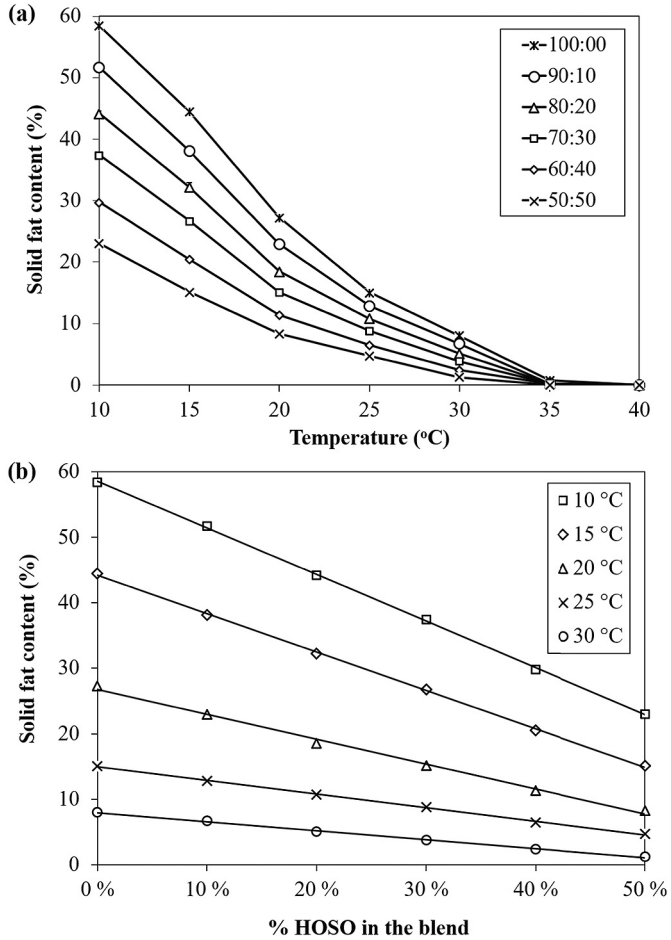

Fig. 1. Solid fat content (%) **(a)** and compatibility diagram **(b)** of anhydrous milk fat (100:00) and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w) at different temperatures. The determination coefficients (R^2) of the compatibility of lipid blends was 0.99 for all formulations at the temperatures studied.

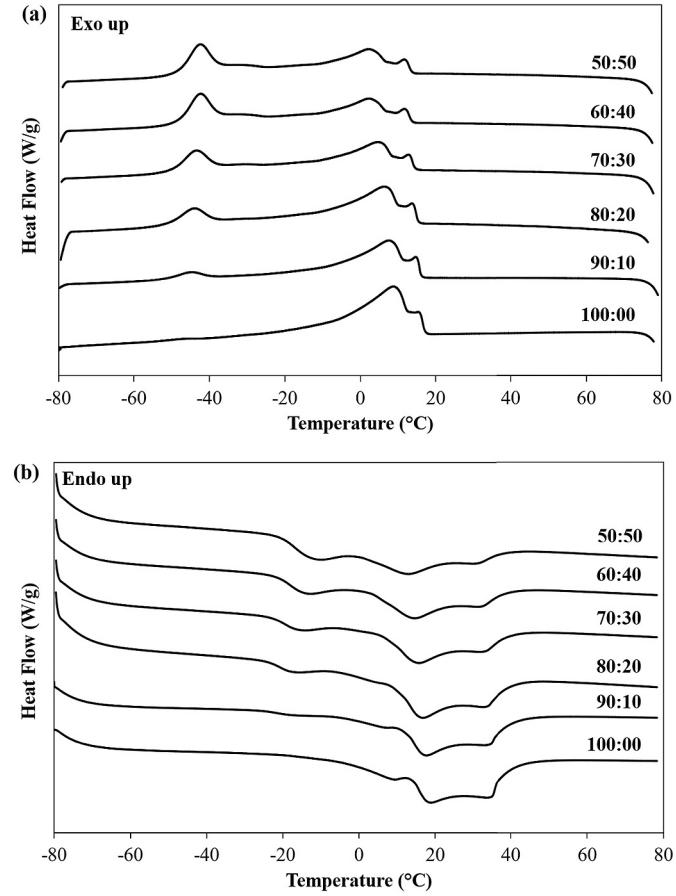

Fig. 2. Thermal behavior of crystallization **(a)** and melting **(b)** of anhydrous milk fat (100:00), and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w).

blends, which ranged from 31.76 ± 0.02 °C to 27.68 ± 0.09 °C for the AMF and 50:50 AMF:HOSO, respectively (shown in the Supplementary material, Table TS2). This reduction was due to both the decrease in the concentration of trisaturated triacylglycerols from AMF, such as SSS and PSS (S = stearic acid, P = palmitic acid), which have a melting point of 65 °C and 61.1 °C respectively, and the increase in triunsaturated triacylglycerols from HOSO, with melting points between −14 °C and 5.6 °C (O'Brien, 2009). The melting points of the blends were in accordance with the applicability requirements of fats for different industrial purposes (Ghotra, Dyal, & Narine, 2002). Particularly, lipid bases with melting points below the body temperature can be used directly as technical fats or shortenings, since they melt completely and do not produce the waxy sensation during consumption (Karabulut et al., 2004).

3.3. Thermal behavior

The AMF presented a wide crystallization range with a single peak initiating at 17.41 ± 0.02 °C and finishing at −39.62 ± 0.22 °C, releasing 68.44 ± 0.69 J/g of energy. Fig. 2a clearly shows two crystallization events. In the first event, the high melting TAGs associate with each other and co-crystallize, while the medium and low melting TAGs crystallize in the second event (Toro-Vazquez, Diblidox-Alvarado, Herrera-Coronado, & Charo-Alonso, 2001). Our results corroborate the findings of Tomaszewska-Gras (2013) and Lambert, Bougrioua, Abbas, Couraivre, & Bresson (2017) who also observed two crystallization events for AMF. The addition of HOSO to the AMF led to the appearance of a second crystallization peak, at −40.09 ± 0.32 °C to −36.55 ± 0.24 °C and finishing at −53.95 ± 2.24 °C to -50.86 ± 0.64 °C (Fig. 2a). The appearance of the second peak was accompanied by the reduction of the first peak, corresponding to the high melting TAGs from the AMF and the increase in enthalpy of the second peak (Table 3). These changes in thermal behavior were probably due to the proportional increase in TAGs concentration from the HOSO containing 54 carbon atoms. This effect is associated with the reduction of the number of molecules involved in the crystallization process, which is proportional to the amount of energy released in the system (Humphrey & Narine, 2004).

Concerning the melting behavior (Fig. 2b and Table 3), the AMF started melting at −16.64 ± 0.31 °C and finished at 45.99 ± 1.78 °C, with release of 68.07 ± 0.73 J/g of energy. A melting peak was observed, which was divided into three temperature ranges, in which the crystals with low and medium melting point are initially melted, followed by the high melting point crystals. Percentages > 20% HOSO added to the AMF resulted in two defined melting peaks. Moreover, although the amount of energy released in the thermal transitions decreased with increasing the HOSO concentrations, it was higher for the second peak when compared to the first one, due to the greater number of molecules involved in the melting process of TAGs. Probably, the first peak corresponds to the triunsaturated and diunsaturated TAGs, which have a lower melting temperature (−15 to 22.8 °C) (O'Brien, 2009), while the second peak is characteristic of medium and high melting fractions of the AMF. For all blends, the highest melting was observed between 15 and 20 °C, which corroborates the reduction of the solid fat content of the lipid bases in this temperature range (Fig. 1a). The

Table 3
Onset temperature (T_o), end temperature (T_{end}), peak temperature (T_p) and enthalpy (ΔH) of crystallization and melting of anhydrous milk fat (100:00) and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w).

	Blend	Peak	T_o (°C)	T_{end} (°C)	T_p (°C)	ΔH (J/g)
Crystallization	100:00	1	17.41 ± 0.02	−39.62 ± 0.22	8.64 ± 0.09	68.44 ± 0.69
		2	–	–	–	–
	90:10	1	17.02 ± 0.33	−25.71 ± 3.59	7.54 ± 0.08	46.66 ± 6.25
		2	−40.09 ± 0.32	−53.95 ± 2.24	−45.45 ± 0.46	2.55 ± 0.57
	80:20	1	15.87 ± 0.08	−8.02 ± 0.48	6.55 ± 0.35	22.89 ± 2.82
		2	−38.63 ± 0.40	−52.94 ± 0.32	−44.59 ± 0.05	6.18 ± 0.09
	70:30	1	15.36 ± 0.24	−12.13 ± 0.46	4.76 ± 0.27	22.71 ± 1.21
		2	−37.79 ± 0.64	−52.03 ± 1.97	−43.84 ± 0.03	8.05 ± 1.10
	60:40	1	14.31 ± 0.79	−13.83 ± 0.81	2.70 ± 0.10	19.35 ± 1.68
		2	−36.68 ± 0.09	−51.71 ± 1.45	−42.83 ± 0.11	11.31 ± 1.10
	50:50	1	13.24 ± 0.00	−13.46 ± 1.12	1.17 ± 0.20	13.45 ± 0.77
		2	−36.55 ± 0.24	−50.86 ± 0.64	−41.15 ± 1.12	14.06 ± 0.29
Melting	100:00	1	−16.64 ± 0.31	45.99 ± 1.78	18.53 ± 0.33	68.07 ± 0.73
		2	–	–	–	–
	90:10	1	−24.49 ± 0.16	45.21 ± 0.64	17.64 ± 0.07	61.80 ± 3.83
		2	–	–	–	–
	80:20	1	−27.20 ± 0.16	−10.02 ± 0.56	−18.82 ± 0.28	2.45 ± 0.27
		2	−5.31 ± 0.64	43.77 ± 0.48	16.64 ± 0.05	49.65 ± 1.87
	70:30	1	−25.07 ± 0.36	−6.79 ± 1.44	−16,68 ± 0.55	4,06 ± 1.11
		2	−2.89 ± 1.78	43.56 ± 0.56	15.93 ± 0.16	35.64 ± 3.15
	60:40	1	−27.36 ± 0.42	−5.04 ± 1.12	−15.33 ± 0.34	5.58 ± 0.79
		2	−1.20 ± 0.18	43.34 ± 0.76	14.53 ± 0.14	33.25 ± 0.21
	50:50	1	−28.64 ± 0.48	−3.31 ± 0.56	−14.04 ± 0.06	6.70 ± 0.01
		2	0.05 ± 0.24	40.17 ± 1.04	13.29 ± 0,02	21.55 ± 0.10

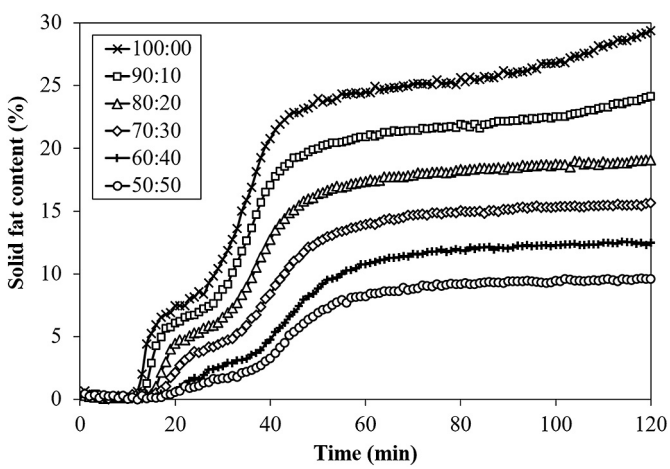

Fig. 3. Isothermal crystallization (15 °C) of anhydrous milk fat (100:00), and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w).

highest melting peak observed at 15 °C (Fig. 2b) corresponds to short chain monounsaturated or trisaturated triacylglycerols, while the last melting point (~33 °C) corresponds to the melting of long chain trisaturated TAGs (Timms, 1980).

When comparing the crystallization and melting thermal behavior (Figs. 2a, b), most of the crystallization occurred at lower temperatures than the corresponding melting temperatures. This result was probably due to the increase in the concentration of liquid triacylglycerols (low melting point) and the consequent reduction of the high melting triacylglycerols in the blends due to the addition of HOSO to the AMF. As a result, a major energy source was required for nucleation and crystal growth of the blends as compared to milk fat (AMF:HOSO 100:00). The same behavior was previously observed in blends made from milk fat and hydrogenated cottonseed oil (Shen et al., 2001) and milk fat stearin and fish oil (Li, Truong, & Bhandari, 2017).

3.4. Crystallization kinetics and microstructure

The crystallization isotherms (Fig. 3) of the AMF and the 90:10, 80:20 and 70:30 AMF:HOSO blends showed two distinct phases of rapid increase of the solid fat content. The first phase was formed after the induction time, which was lower for the AMF (13.00 ± 0.05 min) when compared to the 70:30 blend (17.67 ± 0.58 min), with no significant differences from the other blends. The formation of a plateau was observed, followed by another phase with a rapid increase in the solid fat content before reaching the equilibrium at the maximum crystallization. Blends of milk fat and cocoa butter also showed crystallization in two phases. The first one corresponded to the crystallization of the melted fat in the α-form and the second involved the polymorphic transformation from α to β′ without a direct crystallization from the fat melt in β′ (Foubert, Dewettinck, Janssen, & Vanrolleghem, 2006). It is also observed that the addition of 40 and 50% HOSO to the AMF resulted in crystallization curves with longer

Table 4
Induction period(t_{SFC}), maximum solid fat content ($SFC_{máx}$), Avrami constant (k), Avrami exponent (n), and the respective coefficients of determination of anhydrous milk fat (100:00), and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO %).

Blends	t_{SFC} (min)	$SFC_{máx}$ (%)	k (min^{-1})	n	R^2
100:00	13.00 ± 0.05[d]	29.57 ± 0.28[a]	3.13×10^{-3} ± 0.00[a]	1.54 ± 0.00[d]	0.97
90:10	14.00 ± 0.58[cd]	24.23 ± 0.44[b]	7.01×10^{-6} ± 0.00[b]	3.25 ± 0.01[b]	0.98
80:20	16.00 ± 0.00[cd]	19.13 ± 0.05[c]	6.72×10^{-6} ± 0.00[b]	3.24 ± 0.00[b]	0.99
70:30	17.00 ± 0.58[c]	15.70 ± 0.17[d]	4.93×10^{-6} ± 0.00[c]	3.24 ± 0.00[b]	0.99
60:40	22.00 ± 0.58[b]	12.69 ± 0.16[e]	3.01×10^{-6} ± 0.00[d]	3.27 ± 0.00[a]	0.99
50:50	28.00 ± 3.21[a]	9.71 ± 0.09[f]	4.27×10^{-6} ± 0.00[c]	3.18 ± 0.00[c]	0.99

[(a,b,c,d,e,f)] Different letters in the same column differ significantly by Tukey's test ($p < .05$).

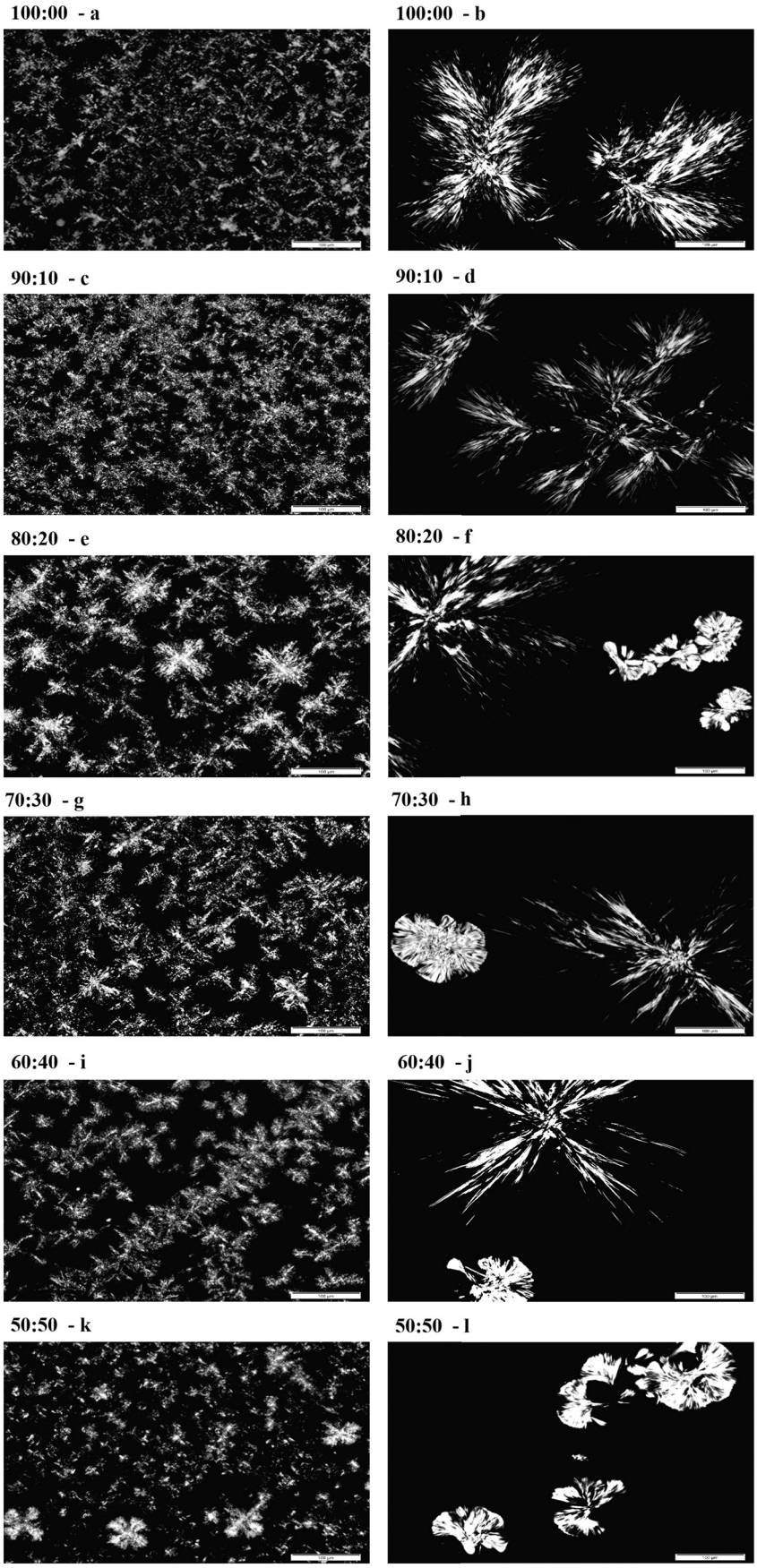

Fig. 4. Microscopy images under polarized light of anhydrous milk fat (100:00), and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w) after stabilization at 15 °C (**a, c, e, g, i, k**) and 30 °C (**b, d, f, h, j, l**). 20× magnification.

Table 5
Parameters of the crystalline network of anhydrous milk fat (100:00), and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO %) at 15 °C.

Blend	Crystaline elements (n°)	Average density	D_{mean} crystals (μm)	% agglomerated crystals	D_{mean} individual crystals (μm)
100:00	13.117	88.00 ± 16.12	11.63 ± 14.19	46.79	5.31
90:10	12.150	63.55 ± 11.16	11.97 ± 18.35	45.90	5.32
80:20	9.605	59.79 ± 9.71	12.45 ± 16.48	47.40	5.38
70:30	10.324	47.97 ± 7.98	13.14 ± 31.38	49.75	5.41
60:40	6.716	49.95 ± 10.25	14.31 ± 16.64	54.91	5.52
50:50	5.634	40.27 ± 6.82	13.83 ± 12.42	59.57	5.83

Crystalline elements: total number of crystals present in an image; Average density: mean density of the crystalline area of an image; D_{mean} crystals: mean crystal diameters; % agglomerated crystals: proportion of agglomerated crystals in an image; D_{mean} individual crystals: mean diameter of non-agglomerated crystals of an image.

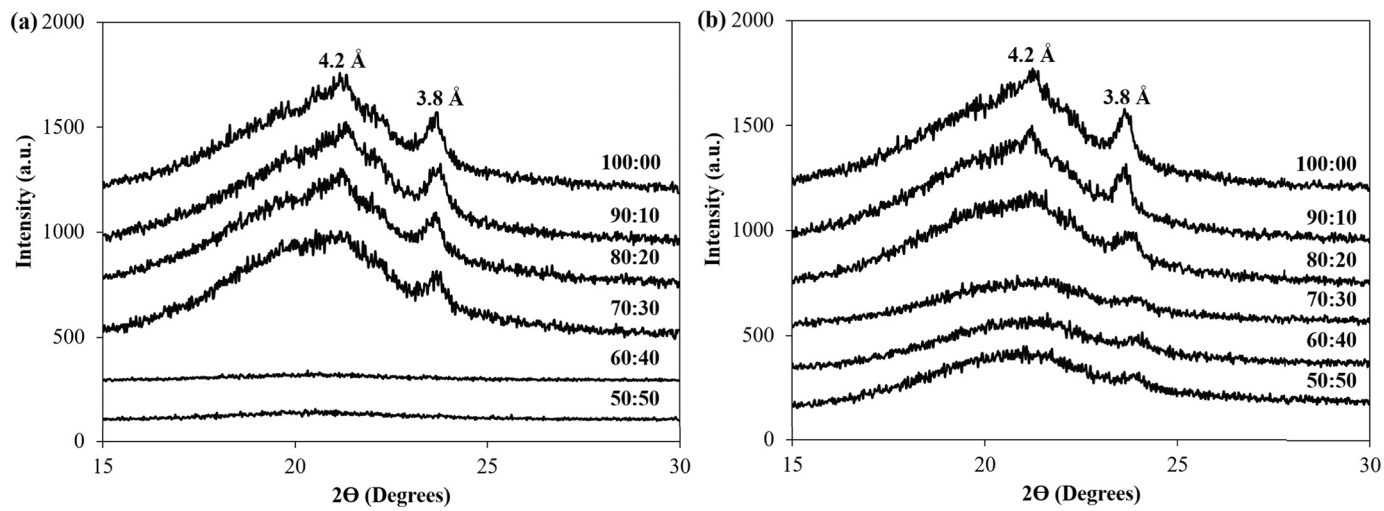

Fig. 5. X-ray diffraction patterns of anhydrous milk fat (100:00), and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w) at 15 °C after 7 **(a)** and 50 days **(b)**.

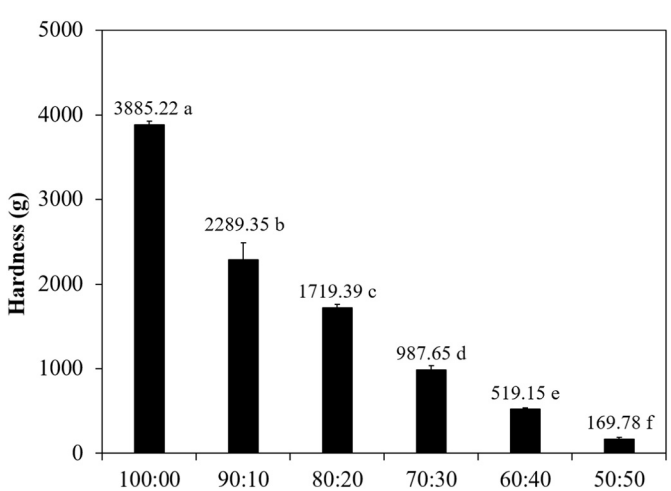

Fig. 6. Hardness of anhydrous milk fat (100:00) and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w) at 15 °C.
(a,b,c,d,e,f) Means with different letters differ significantly from each other by the Tukey's test ($p < .05$).

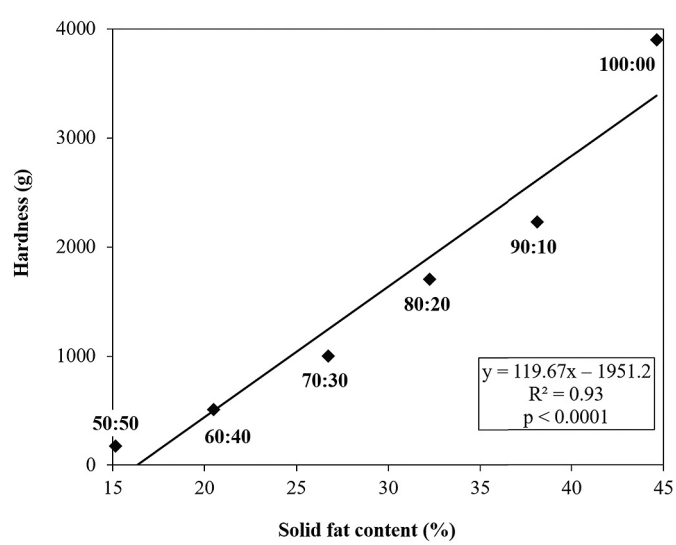

Fig. 7. Correlation between hardness and solid fat content of anhydrous milk fat (100:00) and AMF and high oleic sunflower oil blends (90:10; 80:20; 70:30; 60:40; and 50:50 AMF:HOSO % w/w) at 15 °C.

induction periods probably due to the lower concentration of trisaturated TAGs, which initiates the crystallization process.

As can be seen in Table 4, there was a significant reduction in the maximum solid fat content (SFC_{max}) for all blends and the Avrami constant (k) with the increase in the HOSO ratio due to the dilution of the saturated fatty acids in the AMF. The Avrami (n) exponent, which indicates the crystallization mechanism, ranged from 1 to 2 for the AMF, which suggests a needle-like crystal growth from instantaneous or sporadic nuclei. This behavior can be clearly observed in the microstructure of the AMF at 30 °C (Fig. 4b), with crystals in the form of long and fine needles of the spherulite-type A, arranged minutely in a radial manner. At 15 °C, a dense crystal, with a high number of crystalline

elements was observed (Fig. 4a and Table 5). The addition of HOSO to the AMF changed the crystallization behavior of the blends, and induced the growth of crystals in the form of discs, resulting in a simultaneous formation of crystals with different morphologies, which is clearly observed at 30 °C for the 90:10, 80:20, 70:30, 60:40, and 50:50 AMF:HOSO blends (Figs. 4d, f, h, j, and l, respectively). These results corroborate the crystallization behavior proposed by the Avrami model, which, with *n* values between 3 and 4, suggests a growth mechanism of spherulite-type crystals with instantaneous nucleation or disc type crystals with sporadic nucleation. A reduction in the number of crystalline elements and the average crystal density with a concomitant increase in the amount of liquid oil in the crystal network was observed with the addition of HOSO to the AMF (Table 5 and Fig. 4c, e, g, i, and k).

Both the AMF and the lipid bases from the AMF:HOSO blends had an average crystal diameter lower than 30 μm, and the high standard deviation (Table 5), which is characteristic of crystallized fats observed under the polarized light (Rousseau, Hill, & Marangoni, 1996b; Rodrigues et al., 2003). In addition, the increase in the HOSO ratio in the blends led to the formation of a larger amount of agglomerated crystals (Table 5), suggesting that the software Image Pro Plus may have considered clusters as a single crystal. When only the individual crystals were evaluated, all blends had diameters ranging from 5.31 to 5.83 μm. Both the crystal diameters and the diameter of the individual crystals were lower than 30 μm, which is desirable for a fat to have technological application in foods (Herrera & Rocha, 1996).

3.5. Polymorphism

The diffractograms of the lipid blends after 7 and 50 days of stabilization at 15 °C are presented in Fig. 5a and b. After 7 days, the AMF and the 90:10, 80:20, and 70:30 AMF:HOSO blends showed a crystalline phase with short spacings at 4.2 e 3.8 Å, which is characteristic of the β′ polymorphic form (Ramel & Marangoni, 2017), whereas no crystalline phase was observed for the 60:40 and 50:50 AMF:HOSO blends. After 50 days, the pre-crystallized blends retained the β′ form, while the 60:40 and 50:50 AMF:HOSO blends crystallized in the same polymorphic form (β′). Our results showed that the HOSO, which tends to form crystals in the β-form due the high oleic acid concentration (Acevedo & Marangoni, 2010), resulted in stable blends crystallized in the β′ form after 50 days when mixed with AMF.

Our results suggest that the AMF allowed obtaining lipid bases containing up to 30% HOSO, structured in the β′ form, after 7 days of stabilization, probably due to the palmitic acid levels above 20%, and the high concentration of asymmetric triacylglycerols, such as POO (P = palmitic acid, O = oleic acid), PLO (P = palmitic acid, L = linoleic acid, O = oleic acid), SPO (S = stearic acid, P = palmitic acid, O = oleic acid) and PPS (P = palmitic acid, S = stearic acid), which crystallize preferentially in the polymorphic form β′ (Timms, 1984). Crystallization in the β′ form observed in our blends is desirable for the production of plastic fats since they provide crystals with a smaller diameter, softer fats, and better sensory quality (Ghotra et al., 2002).

3.6. Hardness

The hardness of a semi-solid lipid system depends on both its solid fat content and microstructural factors. The hardness of the lipid bases decreased drastically with the increase in HOSO concentration in the blends (Fig. 6), with a significant correlation with the reduction of solid fat content (Fig. 7). Significant linear relationships between these variables indicate that consistency is strongly dependent on solid fat content and can be used for predictive purposes. Besides being an important aspect of the functionality of plastic fats, this physical property is recognized as an important quality attribute. Although the AMF has a wide melting range, it is considered a hard fat at low temperatures. The modification of the solid fat content and the crystalline network by the incorporation of liquid oils can affect the sensory attributes and the technological performance of lipid bases, guiding their application in food products (Braipson-Danthine & Deroanne, 2004; Piska, Zárubová, Louzecký, Karami, & Filip, 2006).

4. Conclusions

The anhydrous milk fat was able to confer structure to plastic lipid bases with a reduced content of saturated fatty acids, obtained from blends containing AMF with high oleic sunflower oil. All blends were compatible at the different temperatures evaluated, evidencing that high oleic sunflower oil was trapped in the crystal network of milk fat. The crystal network showed crystals in the polymorphic form β′ with a diameter smaller than 30 μm. Plastic fats with this profile are suitable for different technological applications. In short, the lipid system of this study enabled the production of plastic lipid bases structured by milk fat, without the use of chemical modification methods.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodres.2018.05.015.

References

Acevedo, N. C., & Marangoni, A. G. (2010). Characterization of the nanoscale in triacylglycerol crystal networks. *Crystal Growth & Design, 10*, 3327–3333.
Allman-Farinelli, M. A., Gomes, K., Favaloro, E. J., & Petocz, P. (2005). A diet rich in high-oleic-acid sunflower oil favorably alters low-density lipoprotein cholesterol, triglycerides, and factor VII coagulant activity. *Journal of the American Dietetic Association, 105*, 1071–1079.
Antoniosi Filho, N. R., Mendes, O. L., & Lanças, F. M. (1995). Computer prediction of triacylglycerol composition of vegetable oils by HRGC. *Chromatographia, 40*, 557–562.
AOCS (2009). *Official methods and recommended practices of the American Oil Chemists' Society*. Champaign: American Oil Society.
Biliaderis, C. G. (1983). Differential scanning calorimetry in food research – A review. *Food Chemistry, 10*, 239–265.
Braipson-Danthine, S., & Deroanne, C. (2003). Blending of hydrogenated low-erucic acid rapeseed oil, low-erucic acid rapeseed oil, and hydrogenated palm oil or palm oil in the preparation of shortenings. *Journal of the American Oil Chemists' Society, 80*, 1069–1075.
Braipson-Danthine, S., & Deroanne, C. (2004). Influence of SFC, microstucture and polymorphism on texture (hardness) of binary blends of fats involved in the preparation of industrial shortenings. *Food Research International, 37*, 941–948.
Brasil (1996). Ministério da Agricultura, Pecuária e Abastecimento. Portaria n° 146, de 07 de março de 1996. *Regulamento Técnico de Identidade e Qualidade de Gordura Anidra de Leite (ou Butteroil). (Anexo VIII)*Brasília: Diário Oficial da União. Available in: http://www.agricultura.gov.br.
Brasil (2005). Agência Nacional de Vigilância Sanitária. Resolução RDC n° 270, de 22 de setembro de 2005. *Regulamento Técnico para óleos vegetais, gorduras vegetais e creme vegetal*Brasília: Diário Oficial da União. Available in: http://www.agricultura.gov.br.
Brasil (2006). Ministério da Agricultura, Pecuária e Abastecimento. Instrução Normativa n° 49, de 22 de dezembro de 2006. *Regulamento Técnico de Identidade e Qualidade de Óleos Vegetais Refinados. (Anexo I)*Brasília: Diário Oficial da União. Available in: http://www.agricultura.gov.br.
Campos, R. (2005). Experimental methodology. In A. G. Marangoni (Ed.). *Fat Crystal Network* (pp. 267–349). New York: Marcel Dekker.
Christie, W. W. (1982). A simple procedure for rapid transmethylation of glycerolipids and cholesterol esters. *Journal of Lipid Research, 23*, 1072–1075.
Cruz-Hernandez, C., Kramer, J. K. G., Kennelly, J. J., Glimm, D. R., Sorensen, B. M., Okine, E. K., ... Weselake, R. J. (2007). Evaluating the conjugated linoleic acid and trans 18:1 isomers in milk fat of dairy cows fed increasing amounts of sunflower oil and a constant level of fish oil. *Journal of Dairy Science, 90*, 3786–3801.
Danthine, S. (2012). Physicochemical and structural properties of compound dairy fat blends. *Food Research International, 48*, 187–195.
Foubert, I., Dewettinck, K., Janssen, G., & Vanrolleghem, P. A. (2006). Modelling two-step

isothermal fat crystallization. *Journal of Food Engineering, 75*, 551–559.
Gamboa, O., & Gioielli, L. A. (2006). Comportamento de cristalização de lipídios estruturados a partir de gorduras de palmiste e óleo de peixe. *Química Nova, 29*, 646–653.
Ghotra, B., Dyal, S. D., & Narine, S. S. (2002). Lipid shortenings: A review. *Food Research International, 35*, 1015–1048.
Grimaldi, R., Gonçalves, L. A. G., Gioielli, L. A., & Simões, I. S. (2001). Interactions in interesterified palm and palm kernel oil mixtures. I-solid fat content and consistency. *Grasas y Aceites, 52*, 349–354.
Guedes, A. M. M., Ming, C. C., Ribeiro, A. P. B., Silva, R. C., Gioielli, L. A., & Gonçalves, L. A. G. (2014). Physicochemical properties of interesterified blends of fully hydrogenated crambe abyssinica oil and soybean oil. *Journal of the American Oil Chemists' Society, 91*, 111–123.
Hartman, L., & Lago, R. (1973). Rapid preparation of fatty acid methyl esters from lipids. *Laboratory Practice, 22*, 475–476.
Herrera, M. L., & Rocha, F. J. M. (1996). Effects of sucrose ester on the kinetics of polymorphic transition in hydrogenated sunglower oil. *Journal of the American Oil Chemists' Society, 73*, 321–326.
Humphrey, K. L., & Narine, S. S. (2004). A comparison of lipid shortening functionality as a funcion of molecular ensemble and shear: Crystallization and melting. *Food Research International, 37*, 11–27.
Karabulut, I., Turan, S., & Ergin, G. (2004). Effects of chemical interesterification on solid fat content and slip melting point of fat/oil blends. *European Food Research and Technology, 218*, 214–229.
Kramer, J. K. G., Hernandez, M., Cruz-Hernandez, C., Kraft, J., & Dugan, M. E. R. (2008). Combining results of two GC separations partly achieves determination of all cis and trans 16:1, 18:1, 18:2 and 18:3 except CLA isomers of milk fat as demonstrated using ag-ion SPE fractionation. *Lipids, 43*, 259–273.
Lambert, A., Bougrioua, F., Abbas, O., Couraivre, V., & Bresson, S. (2017). Temperature dependent Raman and X-ray diffraction studies of anhydrous milk fat. *Food Chemistry*. http://dx.doi.org/10.1016/j.foodchem.2017.09.006.
Lands, W. E. M. (2005). *Fish, Omega-3, and human health* (2nd ed.). Champaign: AOCS Press.
Li, B. Z., Truong, T., & Bhandari, B. (2017). Crystallization and melting properties of mixtures of milk fat stearin and omega-3 rich oils. *Food Chemistry, 218*, 199–206.
Lida, A. M. D. N., & Ali, A. R. (1998). Physico-chemical characteristics of palm-based oil blends for the production of reduced fat spreads. *Journal of the American Oil Chemists' Society, 75*, 1625–1631.
MacGibbon, A. K. H., & Taylor, M. W. (2006). Composition and structure of bovine milk lipids. In P. F. Fox, & P. L. H. McSweeney (Vol. Eds.), (3rd ed.). *Lipids: Vol. 2. Advanced Dairy Chemistry* (pp. 1–42). New York: Springer.
Martin, C. A., Carapelli, R., Visantainer, J. V., Matsushita, M., & Souza, N. E. (2005). Trans fatty acid content of Brazilian biscuits. *Food Chemistry, 93*, 445–448.
Mba, O. I., Dumont, M. J., & Ngadi, M. (2015). Palm oil: Processing, characterization, and utilization in the food industry – A review. *Food Bioscience, 10*, 26–41.
Mozaffarian, D., Katan, M. B., Ascherio, A., Stampfer, M. J., & Willett, W. C. (2006). Trans fatty acids and cardiovascular disease. *New England Journal of Medicine, 354*, 1601–1613.
O'Brien, R. D. (2009). *Fats and oils: Formulating and processing for applications* (3rd ed.). Boca Raton: CRC Press.
Parodi, P. W. (2009). Milk fat nutrition. In A. Y. Tamime (Ed.). *Dairy fats and related products* (pp. 28–51). New Jersey: Wiley-Blackwell.
Patterson, H. B. W. (2010). Bleaching of Important Fats and Oils. In G. List (Ed.). *Bleaching and Purifying Fats and Oils: Theory and Practice* (pp. 97–151). (2nd ed.). Champaign: AOCS Press.
Piska, I., Zárubová, M., Louzecký, T., Karami, H., & Filip, V. (2006). Properties and crystallization of fat blends. *Journal of Food Engineering, 77*, 433–438.
Quast, L. B., Luccas, V., Ribeiro, A. P. B., Cardoso, L. P., & Kieckbusch, T. G. (2013). Physical properties of tempered mixtures of cocoa butter, CBR, and CBS fats. *International Journal of Food Science and Technology, 48*, 1579–1588.
Queirós, M. S., Grimaldi, R., & Gigante, M. L. (2016). Adittion of olein from milk fat positively affects the firmness of butter. *Food Research International, 84*, 69–75.
Ramel, P. R., & Marangoni, A. G. (2017). Characterization of the polymorphism of milk fat within processed cheese products. *Food Structure, 12*, 15–25.
Ramel, P. R., Peyronel, F., & Marangoni, A. G. (2016). Characterization of the nanoscale structure of milk fat. *Food Chemistry, 203*, 224–230.
Ribeiro, A. P. B., Basso, R. C., Grimaldi, R., Gioielli, L. A., & Gonçalves, L. A. G. (2009). Instrumental methods for the evaluation of interesterified fats. *Food Analytical Methods, 2*, 282–302.
Ribeiro, A. P. B., Basso, R. C., & Kieckbusch, T. G. (2013). Effect of the addition of hardfats on the physical properties of cocoa butter. *European Journal of Lipid Science and Technology, 115*, 301–312.
Ribeiro, A. P. B., Grimaldi, R., Gioielli, L. A., & Gonçalves, L. A. G. (2009). Zero trans from soybean oil and fully hydrogenated soybean oil: Physico-chemical properties and food aplications. *Food Research International, 42*, 401–410.
Ribeiro, A. P. B., Grimaldi, R., Gioielli, L. A., Santos, A. O., Cardoso, L. P., & Gonçalves, L. A. G. (2009). Thermal behavior, microstructure, polymorphism, and crystallization properties of zero trans fats from soybean oil and fully hydrogenated soybean oil. *Food Biophysics, 4*, 106–118.
Rodrigues, J. N., Anton, C., & Gioielli, L. A. (2003). Cristalização de lipídios estruturados obtidos a partir de gordura do leite e óleo de milho. *Revista Brasileira de Ciências Farmacêuticas, 39*, 93–103.
Rodrigues, J. N., & Gioielli, L. A. (2003). Chemical interesterification of milkfat and milkfat-corn oil blends. *Food Research International, 36*, 149–159.
Rodrigues, J. N., Torres, R. P., Mancini-Filho, & Gioielli, L. A. (2007). Physical and chemical properties of milkfat and phytosterol esters blends. *Food Research International, 40*, 748–755.
Rodrigues-Ract, J. N., Otting, L. N., Poltronieri, T. P., Silva, R. C., & Gioielli, L. A. (2010). Comportamento de cristalização de lipídios estruturados obtidos a partir de gordura do leite e óleo de girassol. *Ciência e Tecnologia de Alimentos, 30*, 258–267.
Rousseau, D., Forestiere, K., Hill, A. R., & Marangoni, A. G. (1996). Restructuring butterfat through blending and chemical interesterification. 1. Melting behavior and triacylglycerol modifications. *Journal of the American Oil Chemists' Society, 73*, 963–972.
Rousseau, D., Hill, A. R., & Marangoni, A. G. (1996). Restruturing butterfat Throught blending and chemical interesterification. 2. Microstruture and polymorphism. *Modifications. Journal of American Oil Chemists' Society, 73*, 973–981.
Schenk, H., & Peschar, R. (2004). Understanding the structure of chocolate. *Radiation Physics and Chemistry, 71*, 829–835.
Shen, Z., Birkett, A., Augustin, M. A., Dungey, S., & Versteeg, C. (2001). Melting behavior of blends of milk fat with hydrogenated coconut and cottonseed oils. *Journal of the American Oil Chemists' Society, 78*, 387–394.
Smith, K. W., Bhaggan, K., Talbot, G., & van Malssen, K. F. (2011). Crystallization of fats: Influence of minor components and additives. *Journal of the American Oil Chemists' Society, 88*, 1085–1101.
Timms, R. E. (1980). The phase behavior and polymorphism of milk fat, milk fat fractions and fully hardened milk fat. *Australian Journal of Dairy Technology, 35*, 47–53.
Timms, R. E. (1984). Phase behaviour of fats and their mixtures. *Progress in Lipid Research, 23*, 1–38.
Tomaszewska-Gras, J. (2013). Melting and crystallization DSC profiles of milk fat depending on selected factors. *Journal of Thermal Analysis and Calorimetry, 113*, 199–208.
Toro-Vazquez, J. F., Diblidox-Alvarado, E., Herrera-Coronado, H., & Charo-Alonso, M. (2001). Triacylglyceride crystallization in vegetable oils: application of models, measurements, and limitations. In N. Widlak, R. W. Hartel, & S. Narine (Eds.). *Crystallization and solidification properties of lipids* (pp. 53–78). Champaign, IL, USA: AOCS Press.
Truong, T., Morgan, G. P., Bansal, N., Palmer, M., & Bhandari, B. (2015). Crystal structures and morphologies of fractionated milk fat in nanoemulsions. *Food Chemistry, 171*, 157–167.
Wassel, P., & Young, N. W. G. (2007). Food applications of trans fatty acids substitutes. *International Journal of Food Science and Technology, 42*, 503–517.
Wolff, R. L., Bayard, C. C., & Fabein, R. J. (1995). Evaluation of sequential methods for the determination of butterfat fatty acid composition with emphasis on trans-18:1 acids. Application to the study of seasonal variations in French butter. *Journal of the American Oil Chemists' Society, 72*, 1471–1483.
Wright, A. J., Hartel, R. W., Narine, S. S., & Marangoni, A. G. (2000). The effect of minor components on milk fat crystallization. *Journal of the American Oil Chemists' Society, 77*, 463–466.